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Economic evaluation of hybrid electrical systems for rural electrification: A case study of a rural community in Nigeria

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ABSTRACT

Diesel generators have long been in use in rural communities in Nigeria despite the impact of emissions caused by these generators on the environment. Due to the high cost and fluctuation in the diesel price and the need for a greener environment, such electric generating systems seem not to be economical and environmentally friendly and need to be addressed. The potential of utilizing the major renewable energy sources prevailing in the Umuejim Amorka community was investigated based on related site surveys and data collected from each household. This paper also investigates the feasibility and economic viability of employing a hybrid-electric system in rural communities. The HOMER software that analyzes system configuration was used to study the application and functional limitations of each hybridized arrangement. The results showed that the renewable energy (RE) based system, photovoltaic (PV)/battery(B), has the lowest cost of energy (COE) and net profit cost (NPC) compared to the PV/Diesel Generator (DG)/B and standalone DG systems. Although the diesel generator (DG) hybrid-electric system (HES) has lower NPC and COE values compared to PV/DG/B, the DG system has the highest significant pollution emission. Furthermore, the result revealed that the PV/B system has the lowest capital and total costs compared to the other two HESs. Moreso, the PV/B system, has the highest return on investment (ROI), making the system the most economically viable and adjudged to be a better candidate for rural community electrification demands.

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1. Introduction

Electricity is indispensable for the economic growth and social development of a nation. Due to population growth and human development, an estimate of over 3 billion people globally living in rural communities is not connected to the national grid. Apart from grid unavailability in certain rural communities, more than 72% of the world's energy consumption comes from fossil fuels (Ahiduzzaman and Islam 2011). The usage of fossil fuel has been shown to have contributed to global warming, and its inaccessibility is increasing (Ahiduzzaman and Islam 2011; Kang et al. 2019). Furthermore, the supply of fossil fuel is likely to be limited in supply shortly due to government policies, production, and logistic challenges. Thus, efforts are required to develop sustainable alternative energy sources that can address the high global energy demand. Nigeria is a developing country with over 170 million people in the population (Tartiyus, Dauda, and Peter 2015). Approximately 40% of the population are living in the villages and are not connected to the National grid (Elusakin Julius, Olufemi, and Chuks 2014). Suresh et al. (Suresh and Manoharan 2014) reported that the connection of the national grid and supply of electricity to the rural

communities can cause enormous financial obligations of the developing country, thus making independent electricity generation prevalent in the rural communities (Ohijeagbon and Ajayi 2014). Olatomiwa et al. (Olatomiwa, Mekhilef, and Ohunakin 2016) revealed that power sectors have been privatized and a framework was setup to improve the energy supply as well as cope with the fast-growing Nigerian population. The authors recommended the implementation of off-grid power generation for rural communities due to the cost associated with grid connections.

Surrounded by the various renewable energy sources prevailing in the eastern part of Nigeria, solar and wind energy sources are found to be highly abundant and available (Akpan, Isihak, and Udoakah 2013). It was reported that the availability of solar radiation in the country has excellent and viable potential for large-scale solar photovoltaic (PV) applications (Olatomiwa, Mekhilef, and Ohunakin 2016). Despite the existing application of wind and PV solar energy sources in different areas of the country, fluctuation in the energy supply is often experienced and requires a combination of adequate and appropriate hybrid electric components (Olatomiwa et al. 2015). Encumbered by the fluctuation in

the energy supply, Olatomiwa et al. (Olatomiwa et al. 2015; Olatomiwa, Mekhilef, and Ohunakin 2016) reported that Nigeria, specifically the eastern part of the country, is a poor wind regime based on meteorological data and may not provide the required wind speed. In an attempt to address the rural electricity demand, researchers have performed several studies to understand alternative electricity sources and systems for the electrification of rural communities (Aziz et al. 2019; Benavente et al. 2019). Different hybrid electric systems (HESs) have been configured and investigated. The influence of hybridized systems in meeting the electricity demands in different communities and cities has also been studied and recommendations made (Bhandari et al. 2014; Ma, Yang, and Lu 2015; Ruth et al. 2014). Laith et al. (Halabi et al. 2017) recommended that social, technical, and institutional factors need to be considered when designing a hybrid energy system to obtain the desired output. A review of the literature shows that the level of research conducted on HESs in Nigeria is limited (Adaramola, Paul, and Oyewola 2014; Olatomiwa et al. 2015). In many cases, the HESs with battery and photovoltaic (PV) components have been used due to their non-depletable and non-polluting nature (Kaabeche and Bakelli 2019). Interestingly, solar energy performance depends on several environmental factors of which photovoltaic (PV) module plays an important role.

The combination of multiple energy sources can not only provide a stable and reliable energy supply but can also reduce the emission from some of the energy sources, making such a system more efficient and cost-effective (Barhoumi et al. 2019; Okonkwo et al. 2020a, 2020b). The PV module is used in the hybrid energy system (HES) and can be arranged as standalone or hybrid connected systems (Belhaouas et al. 2017). In other cases, diesel generators have been added to the HES and have been reported to increase the overall operating cost of the HES (Gebrehiwot et al. 2019). Furthermore, fuel transportation to remote communities can be challenging for diesel-based HES. Importantly, in rural communities where diesel generators are predominantly used due to off-grid connection, combining diesel generator sources with renewable energy sources can boost energy during low-energy supply periods and reduce the effect of rising fuel prices and scarcity. Mehrpoova et al. (Mehrpooya, Mohammadi, and Ahmadi 2018) investigated the economic and environmental impacts of using HES for rural electrification. The results show evidence of optimal application of the HES to rural community electrification. However, some key indicators were not investigated. Nandi et al. (Nandi and Ghosh 2009) studied the option of employing HES and a stand-alone generator for rural electrification in Bangladesh. Different economic indicators were considered in the study. The result showed that COE and NPC decreased significantly for HES application compared to stand-alone diesel generator, presenting the HES as a better candidate for rural electrification.

Different pieces of software have been used for some rural off-grid areas, which has proven to be economical and efficient in the assessment of the techno-economic feasibility of utilizing a hybrid energy system (Halabi et al. 2017; Olatomiwa, Mekhilef, and Ohunakin 2016).

Several authors have studied the viability assessment of different renewable energy sources with HOMER software (Fazelpour, Soltani, and Rosen 2016; Ramli, Hiendro, and Twaha 2015). Lorafe et al. (Lozano et al. 2019) performed a detailed comparative study of off-grid island communities using HOMER software. Two scenarios of HES and PV stand-alone systems were considered. The result showed that HES is more feasible and effective than the stand-alone solar system. HOMER software has been extensively used for off-grid and grid-connected rural community electrification (Amutha and Rajini 2016; Rajbongshi, Borgohain, and Mahapatra 2017). It was reported that HOMER is widely used due to its renewable energy-based system and can provide optimization analysis with limited input, making the HOMER software a potential tool for hybrid energy system feasibility assessment. While HOMER software has been used to investigate the feasibility of rural electrification in different communities and countries, employing this technology in a hybrid setting of the Umuejim Amorka rural community is yet to be explored, providing the novelty of this study. This study bridges this gap and promises to be a game-changer in the efforts to provide a sustainable, eco-friendly power supply to this rural community. This study also seeks to evaluate and recommend the best HES that meets the rural community electrification needs.

The present study also examines the feasibility and application of available energy sources in the Umuejim Amorka community. The novel employment of HOMER software in providing adequate economical analysis and solutions to address the urgent electric power needs of the community that has based its electric generation on DG not only will benefit the community but also can be employed by other rural communities facing similar electric power challenges. The versatility of the HOMER software enhanced the optimal, sizing and performance analysis of the proposed HES in relation to the community and environmental condition. Comparison is made between different hybrid electrical system arrangements. The impacts of such a project on the socio-economic wellbeing of the Umuejim Amorka community have not been reported in the literature. One of the benefits of the study is for the authority and policymakers to decide on electrification of the community with a similar hybrid energy arrangement that provides cost-effectiveness and high efficiency. The versatility of the HOMER software enhanced the optimal, sizing and performance analysis of the proposed HES in relation to the community and environmental conditions.

1.1. Location of the study

Umuejim Amorka is a rural community situated in the eastern part of Nigeria, approximately 1100 km away from the capital city of Abuja. It lies between 5°43.9'N latitudes and 6°52.4'E longitudes on the map as shown in Figure 1.

The global radiation in the community is relatively high throughout the year and the average temperature in the mid-'20s, making the community potential for renewable energy (Olatomiwa, Mekhilef, and Ohunakin 2016). The country's power sector is unreliable and vulnerable due to politics and fuel crises that have affected connecting several rural communities, such as Umuejim Amorka into the grid. Furthermore,

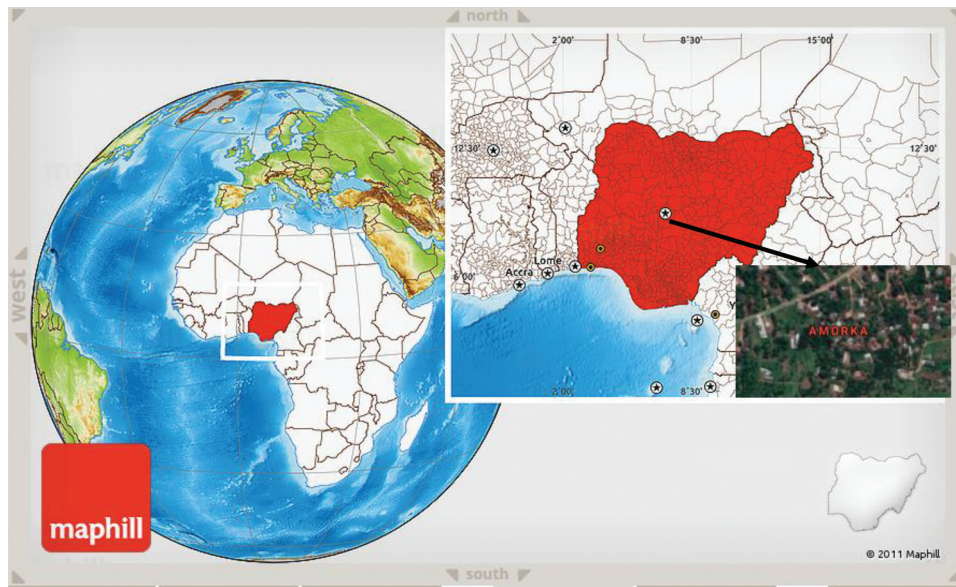


Figure 1. Map of the Umuejim Amorka rural community extracted from HOMER Pro (Energy 2016).

connection to the grid requires a huge amount of investment, which can be challenging for rural communities. As a result, many households have engaged in using diesel generator (DG) as a source of electricity, which is neither economical nor environmentally friendly. The well-defined flow-chart and schematic system configuration of the economic feasibility analysis of the hybrid electric system (HES) for the Umuejim Amorka community electrification are shown in Figure 2. At the inception of the feasible study, different rural areas of the community were visited, and the meteorological and load data of the locations were analyzed.

Based on the outcome of the investigation and examination of the collected data, two possible energy sources were identified available to address the electricity challenges of the Umuejim Amorka community.

2. Methodology

2.1. Description of the system components

The methodology of the HES study is discussed for feasibility and assessment. The system components and descriptions are presented in this section. Scientific expression of the economic models used in the assessment of the feasibility of the HESs is also reported in this section.

2.1.1. Converter

A generic bi-directional converter system is linked between an alternating current (AC) and direct current

(DC) terminals. In HES, the converter converts the AC loads into DC. The efficiency of the inverter is 95%, while the lifetime of the converter is 15 years (Mirhassani et al. 2015). HOMER software performs the sizing of the inverter by optimizing the power flow of the PV, battery, and fuel cell (Energy 2016).

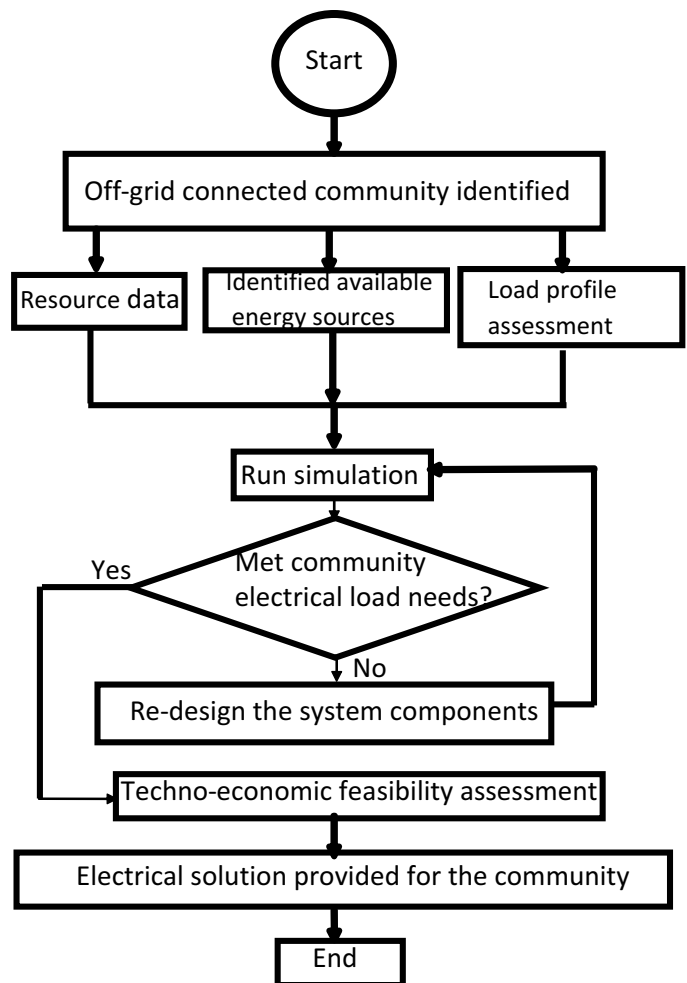


Figure 2. Flowchart of the feasibility study.

2.1.2. PV model

In HES, solar cells are organized in series or parallel in the form of panels to produce power from PV. The amount of electricity produced by this arrangement depends on the efficiency and arrangement of the PV system. The size of the PV and its output are determined to supply the required peak load demand. A 1 KW rated generic flat-plate PV is used in this study, and the cost of the PV panels is taken from the manufacturer's website (Choice 2000).

2.1.3. Diesel genset

A 10 KW generic diesel generator set is used in this study. The primary function of the DG is to supply steady power that not only charges the battery energy storage system but also supplies the load. The selection of generators is based on the daily average maximum load usage.

2.1.4. Battery storage

In a hybrid configuration, during a low supply of renewable energy (RE) sources, energy stored in the battery can support the steady electric energy supply (Prakash, Mustain, and Kohl 2009). The battery bank achieves its stored capacity when a maximum state of charge (100%) is reached through the surplus energy generated by the renewable and non-RE. In this study, a Generic 100 kWh Li-Ion battery with a nominal capacity of 100 kWh and a nominal bus voltage of 600 V was used. The economic and technical characteristics of the selected Generic 100 kWh Li-Ion battery are adopted from HOMER energy software (Energy 2016).

2.2. System description

2.2.1. Load profiles estimation

Assessment and exploration of the Umuejim Amorka community load requirement identified in Figure 2 are important to be able to provide worthwhile options for the community to

achieve their electricity demand objective. The estimated number of households in the selected community is 108. In a typical household, 3–5 LED bulbs are used for lighting, as well as one or two ceiling fans, a television, and phone charging electrical ports. Due to the off-grid connection of the community and non-uniformity in each household's load demand, the exact load profile cannot be ascertained, but estimation was made on-site visitation. Adeoti et al. (Adeoti, Oyewole, and Adegboyega 2001) reported an average electricity consumption of about 2.3 kWh/day for a typical household in a rural community in South West Nigeria. However, the average electricity demands will likely change in the communities due to the instability of electricity and higher fuel price in different parts of Nigeria.

Usually, either of the two methods, day-to-day or step-to-time-step randomness, can be followed by HOMER software in calculating the load profile (Energy 2016). In this study, step-to-time-step randomness of 10% is used, and the result is shown in Figure 3 and 4.

Figure 3 and 4. show the load profile distribution for 24 hours and a given seasonal year generated by HOMER software. According to Figure 3, the minimum load demand occurs between 0.0 am and 4 am (Figure 3). The peak primary load demand is predicted to be between 17:00 and 22:00. To accommodate this variation in load, a variable that considers 20% day-to-day and 15% hour-to-hour step change is incorporated into the HOMER algorithm. The algorithm determines the operating hours of the electrical appliances using a probabilistic approach (Bahramara, Moghaddam, and Haghifam 2016). The appliances and load consumption for the Amorka community under consideration are calculated using Equation 1 and displayed in Table 1 bearing in mind the seasonal load demand.

$$\text{Energy consumption perday(kWh)} = W \times N_{\text{xt}}(h)1000 \quad (1)$$

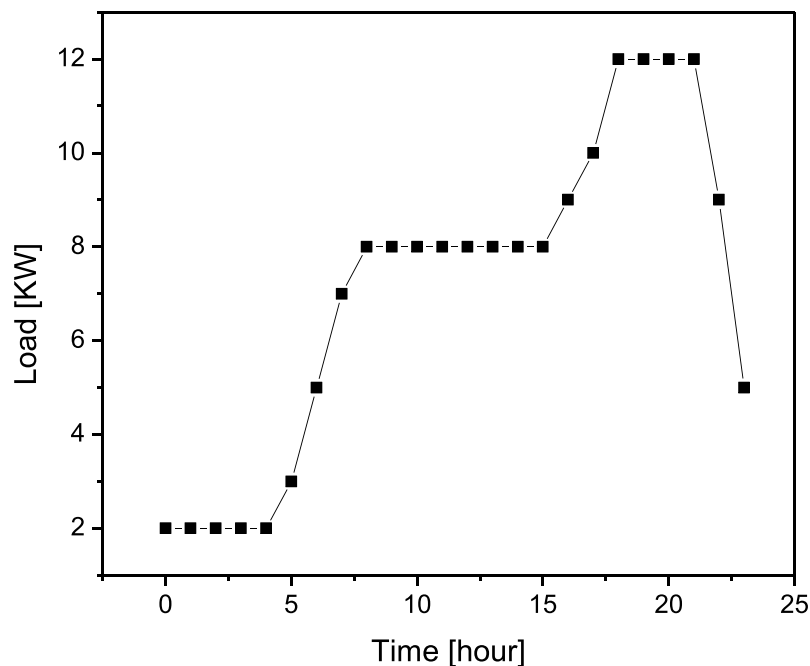


Figure 3. Hourly load consumption profile.

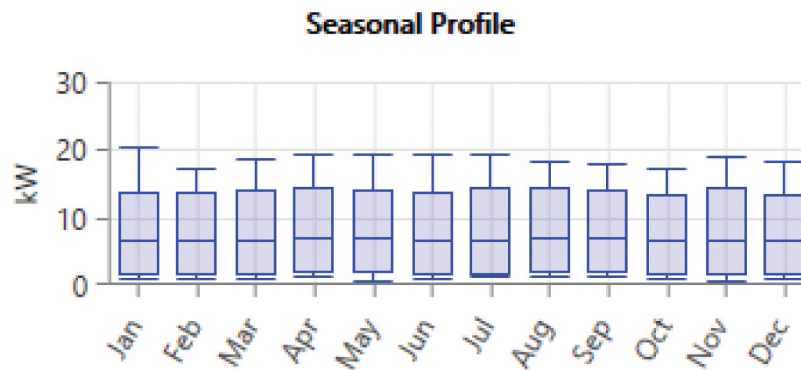


Figure 4. Yearly load consumption profile.

Table 1. Estimated electricity requirements for households in the community.

No	Load type	Quantity required	Power (Watts)	Dry season (Feb.-May.) Working time(h)	Raining season (June-Sept) Working time(h)	Harmattan season (Oct.- Jan) Working time(h)
1	Domestic					
	Bulbs	240	20	8	8	8
	Television	40	80	6	6	6
	Ceiling fans	150	30	12	6	0
	Phone charger	20	10	4	4	4
	Refrigerator	12	120	12	6	4
2	Industrial/commercial/community					
	Shops	15	300	6	6	6
	School	2	400	8	8	8
	Bore hole pump	2	700	4	2	4
	Energy demand/day (Wh)			168,680	130,240	103,160

where

W stands for the appliance rating,

N means the number of appliances, and t represents the operation time.

2.2.2. Solar energy availability and temperature

It is known that the annual solar radiation cycle varies from season to season and the change in the weather is influenced by several factors such as rain, fog, wind, temperature, sunlight (Stine and Huybers 2012). As reported earlier in the study, the Umuejim Amorka community is in an area with high average solar radiation but changes in seasonal weather. Knowledge of index clearness is also required to access the availability and potential of sunlight of the community. The average daily global solar radiation (SR) data of the community was obtained from the NASA Surface Meteorology and solar energy website (NASA) and shown in Figure 5.

HOMER technology calculates the PV array in terms of rated KW and assumes the solar output of the PV array to be linearly proportional to the incident radiation (Seo et al. 2010). In the calculation of the generated electricity, HOMER considers the number of units, capital costs, replacement, and operating hours in the analysis. Hence, a debating factor of 90% is often used by HOMER in the calculation of PV output

and used in the present study (Kansara and Parekh 2011). The information from the latitude and longitude position of a plane is used by HOMER to acquire the clearness index. The clearness index is described as a fraction gained as the result of sending solar radiation to the atmosphere to strike the PV arrays on the earth. Figure 5 displays the daily radiation and clearness index outputs related to the solar radiation for the community.

Monthly variation in the global solar radiation, which influences the output from solar energy conversion systems, is seen in Figure 4 and can be attributed to changes in the annual weather of the community. The relatively low solar radiation seen between April and October is typical of the rainy season dominated by heavy rain clouds (Adaramola, Paul, and Oyewola 2014). The monthly mean solar radiation in the other months of dry and Harmattan seasons shows clear and higher average solar radiation of 5.17 KWh/m²/day in the community.

The functional efficiency of renewable energy sources strongly depends on the predominant environmental conditions. HOMER software determines the average temperature of a location in calculating the PV power efficiency (Praveena, Rao, and Rao 2020). Hence, a clear correlation can be seen between the average solar radiation data shown in Figure 5.

2.3. Economic mathematical models

As highlighted earlier, during the feasibility study, the Umuejim Amorka community was visited as part of the project plan and the meteorological data, as well as loads in different locations of the community, were obtained and analyzed. The survey showed that renewable energy sources (wind and solar energy systems) can be used to address the electric energy challenges in the community despite the potential of high emission standalone generators being used in most households (Dufo-López et al. 2011). The information generated from the feasibility study was used in the hybrid system configuration. The following HESs were configured and analyzed. They include; PV/Diesel Generator/Wind turbine/Battery (PV/DG/B), PV/Battery (PV/B), and Diesel Generator (DG). Evaluation of each HES is performed to understand the suitability of a given system to address the electricity need of the community. This process involves considering some financial parameters such as operating cost, Net Present Cost (NPC), Cost of

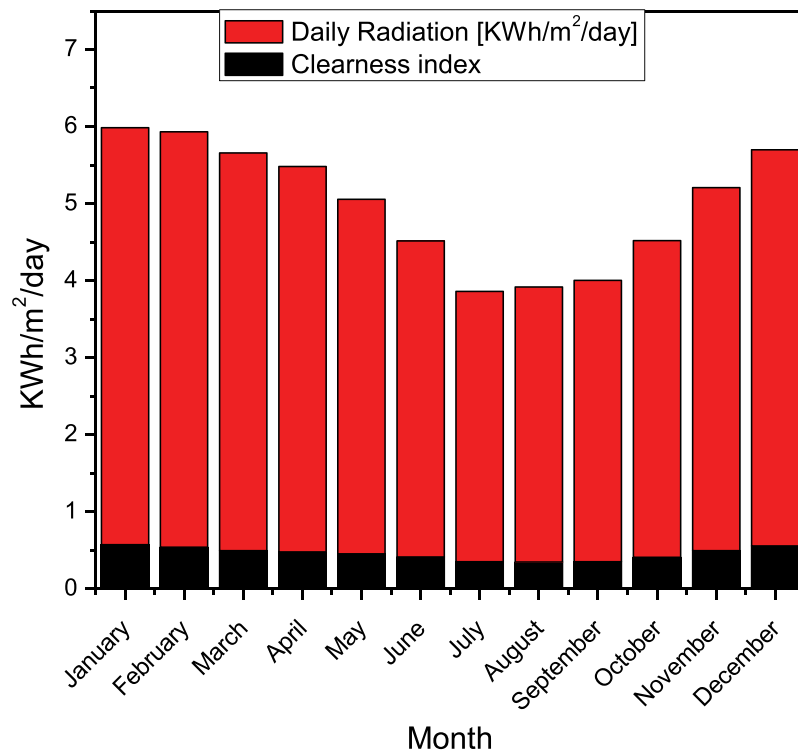


Figure 5. Average solar radiation data of the rural community.

Energy (COE), initial capital cost, and replacement cost of each system. Also, CO₂ emissions, consumption of diesel, production of energy, and excess energy are considered in the economic analysis. During the simulation, HOMER compares the economics of a wide range of system configurations of renewable and non-RE sources, which accounts for the capital and operating cost of each energy source (Liu et al. 2011). HOMER calculates the operating cost of HES as the yearly value of all costs other than the initial capital cost using Equation 2:

$$C_{\text{operating}} = C_{\text{ann.tot}} - C_{\text{ann.cap}} \quad (2)$$

Where:

$C_{\text{ann.tot}}$ = the total yearly cost (\$/year) and $C_{\text{ann.cap}}$ is the total yearly capital cost (\$/year).

The NPC of a HES is used to describe the life cycle cost of a HES in HOMER software. The NPC is calculated by subtracting the values of all the revenues earned by a system from the present value of all the costs incurred over the system's lifetime calculated using Equation 3 (Das et al. 2017).

$$C_{\text{NPC}} = \text{TACCRF}(i, N) \quad (3)$$

Where:

TAC = the total yearly cost (\$/year) and i = the yearly interest rate, N stands for the project lifetime. CRF is the capital recovery factor given as:

$$\text{CRF}(i, N) = i(1+i)^N(1+i)^N - 1 \quad (4)$$

The COE is the average cost per kilowatt-hour of useful electric energy produced by a renewable or non-RE source. It is calculated using the HOMER software using Equation 5 (Das et al. 2017).

$$\text{COE} = \text{TACE}_{\text{anload served}} \quad (5)$$

where $E_{\text{anload served}}$ = the total yearly load served by the system in KWh.

In performing feasibility analysis of systems, alternative economic performance measures such as return on investment (ROI) are considered. In the economic analysis, the feasibility study is also determined by ROI and represented in Equation 6 (Reddy 1997).

$$\text{ROI} = \sum_{t=0}^N C_{i, \text{ref}} - C_{iN}(C_{\text{cap}} - C_{\text{capref}}) \quad (6)$$

Where:

$C_{i, \text{ref}}$ = the reference system nominal yearly cash flow

C_i implies the current system nominal yearly cash flow

C_{cap} is the current capital cost and C_{capref} is the reference capital cost.

In this study, the above four basic economic parameters (NPC, COE, ROI, and IRR) are used to evaluate the performance of the configured HESs.

3. Hybrid system optimization and modeling

HES involves two or more energy sources and a storage system combined to produce a higher efficient source of power than the individual sources. The HES is designed in such a way that the excess electricity produced by the PV or wind sources is stored in the battery (Dolatabadi, Ebadi, and Mohammadi-Ivatloo 2019). The problem of incoherence in using renewable sources in HESs could be eradicated by employing an energy battery storage element. The power generation can be either from renewable or non-renewable energy sources depending

Table 2. System configuration/architecture.

No	Component	Type
1	Generator	Autosize Genset
2	PV	Generic flat-plate PV
3	Storage	Generic 100 KW Li-ion
4	System converter	Converter

on the HES arrangement. By combining different electric energy sources, the disadvantages of each of the components can be minimized (Nelson, Nehrir, and Wang 2006; Zhou et al. 2010). The present study system consists of a photovoltaic, wind turbine, diesel generator with the battery as a backup and storage system as shown in Table 2.

Each of the sources has its own merits and demerits and the application of each source depends on the use and economic viability. Based on the economic assessment, the best system is chosen. Figure 6 displays the schematic diagram of the HESs considered in this study.

In this study, a HOMER technology developed by Natural Renewable Energy Laboratory (NREL) was used in the simulation and techno-economic assessment. Meteorological data of the components is required by HOMER for the simulation and optimization analysis (Shiroudi et al. 2012). The system components described previously in section 2.0 and their designs are incorporated in the HOMER analysis. HOMER also performs economic modeling by decreasing the NPC and COE of the system based on the calculated component annual total cost (Shiroudi et al. 2012). The annualized cost includes capital cost, operation, and maintenance (O&M) cost, replacement minus any miscellaneous expenses, such as CO₂ emission (Li et al. 2020). The annual

interest rate is fixed at 10%. The simulation was performed to obtain the optimal power requirement that meets the electrical load needs of the community.

3.1. Input Data

To optimize the sizing of the various components used in the analysis, the parameters and their respective costs shown in Table 3 are loaded into the HOMER Pro software.

4. Results and discussion

The main objective of this study was to assess the feasibility and economic viability of utilizing the most efficient and effective HES to meet the electrical load demands of the Umuejim Amorka community. Here, HOMER calculates an hourly time series simulation for each possible HES to assess the operational characteristics and economic viability. The simulation results for the various HESs are discussed in the subsequent sections.

4.1. Average monthly electricity production

In Figure 10, the monthly average electricity production of the three HESs is shown. It is evident that the electricity production of pure RE sources is higher than the diesel-based sources and can be attributed to the influence of the community environment on the pure RE-based systems (Olatomiwa et al. 2015). The excess electricity produced by the RE sources is stored in the battery. Figure 7(a) displays the electric production throughout the year for PV/DG/B HES. It can be seen that

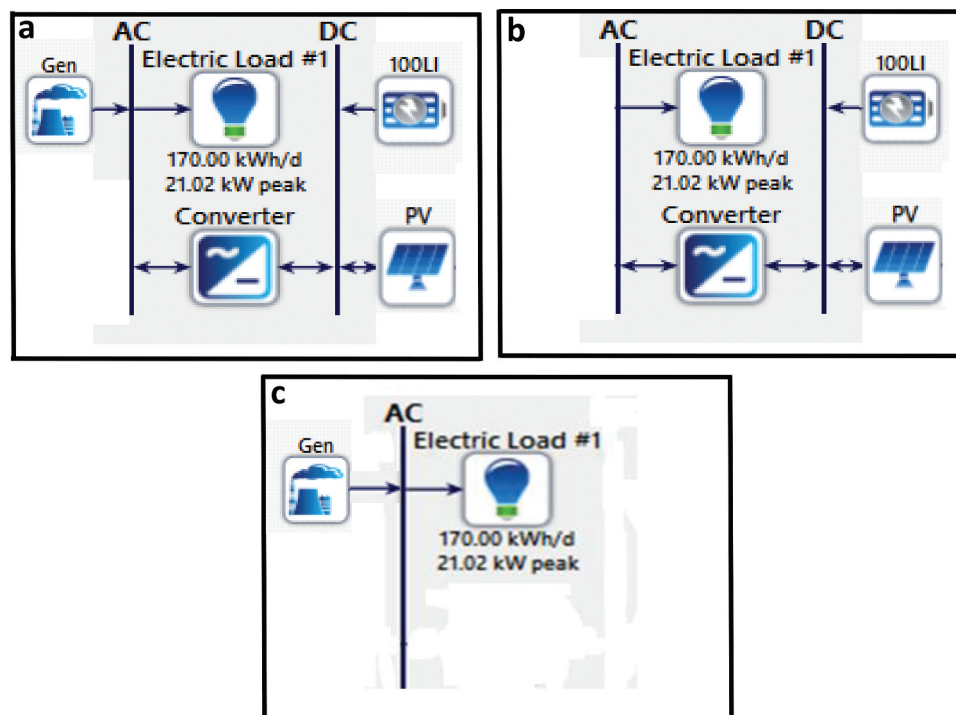
**Figure 6.** Schematic diagram of the hybridized (a) PV/DG/B (b) PV/B (c) DG systems.

Table 3. Input data for simulation in HOMER.

System architecture	Capital cost (\$/KW)
Battery	2,975
PV Modules	1,800
Converter	146
Diesel generator	2,130

while 95.2% of the electricity is produced by the DG, 4.81% is supplied by the PV, underlining the advantages of including PV arrays in enhancing the HES performance.

Figure 7(b) shows a decrease in RE-based electricity production between April and October and is ascribed to the rainy season as reported by Adaramola et al. (Adaramola, Paul, and Oyewola 2014). This is expected because RE sources are mostly influenced by the meteorological conditions in a given location, resulting in occasional disequilibrium between electricity generation and demand that is supplied by stored electricity in the HES (Alihodzic et al. 2014). Figure 7c shows that for the standalone DG, 100% of the electric production is supplied by the DG HES. While the RE-based system (PV/B) shows that the PV supplied

100% of the electricity, the electric production in the DG HES is primarily supplied by a generator, highlighting the influence and contribution of PV in the RE system set up as shown in Figure 7.

4.2. Cost analysis

The cost implications and breakdown of each component in the HES are presented in Table 4. It can be seen that the capital costs for the PV/DG/B and DG HESs are higher than other HES (PV/B) and are ascribed to the high capital conglomerate of the hybrid components that make up the HESs. Furthermore, a gradual reduction in the PV price can be influential to the low capital cost of PV/B HES compared to other HESs as displayed in Table 4. However, these two high capital-cost HES systems have lower replacement and O&M costs than the purely RE-based ones. The average lifetimes of different components of the power system are mentioned in Table 4. Indeed, the lifetime of batteries is 2 years. However, the lifetimes of PV and DG are 20 years and 15 years, respectively. The total cost includes the replacement cost and the salvage value at the end of each period.

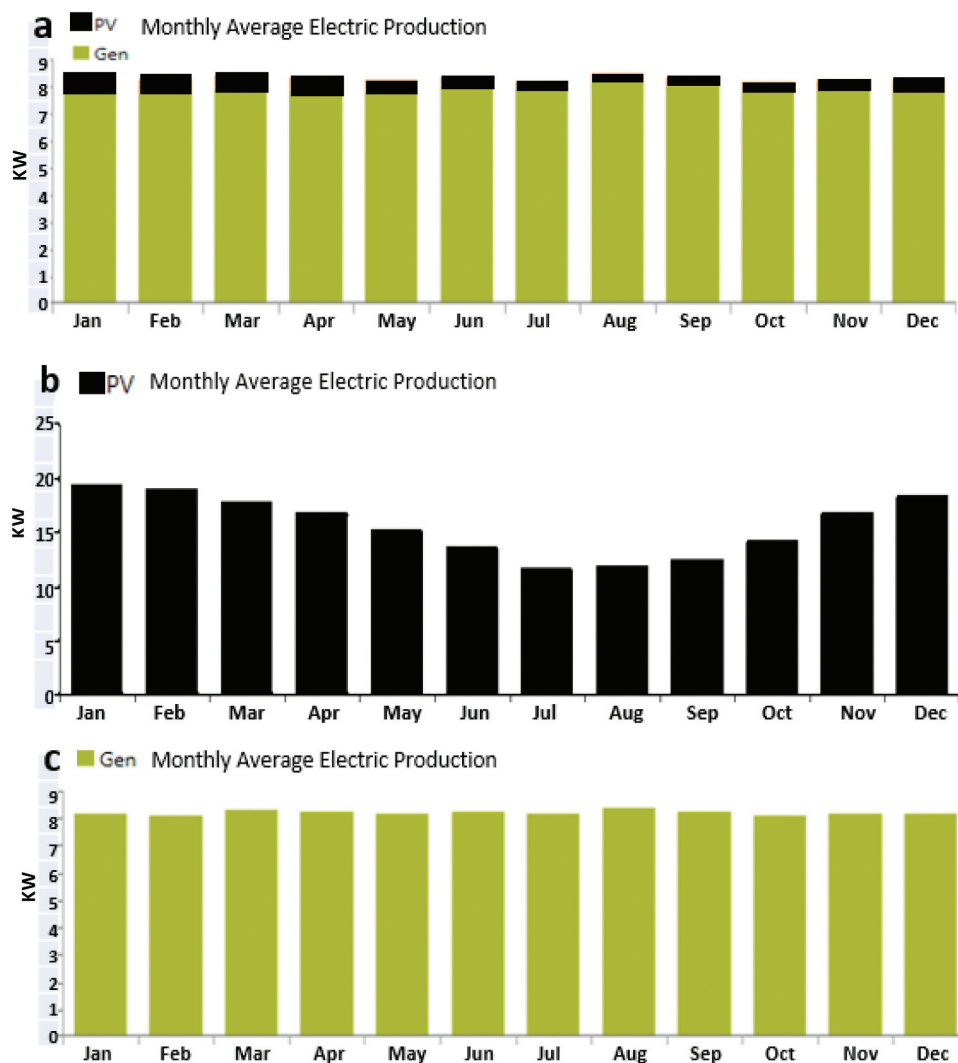
**Figure 7.** Monthly average electric production for (a) PV/DG//B, (b) PV/B (c) DG.

Table 4. Detailed cost of the HESs.

HES	Capital (\$)	Replacement (\$)	Life time (years)	O&M (\$)	Fuel (\$)	Salvage (\$)	Total (\$)
PV/DG/B	462,659.02	38,834.71	25	12,711.27	140,820.10	(6,530.71)	648,494.39
DG	455,350.07	37,689.93	15	11,165.50	144,422.05	(5,819.24)	642,808.31
PV/B	12,000.00	71,727.91	25 years	67,327.95	0.00	(700.89)	150,354.97

The DG-based systems have high fuel costs as well as replacement costs and can be attributed to increasing diesel consumption and sizing of the system components. As can be seen in Table 4, the total cost for the PV/B system is \$150,354.97 which is more than four times larger than the PV/DG/B and standalone DG system. The PV/B system displays a distinguishing difference in the total cost compared to the other two hybrid systems, as is attributed to the PV's low price and the system not utilizing fuel. However, Laith et al. (Halabi et al. 2017) reported that the high replacement cost observed in the PV/B system is due to high battery replacement costs.

Figure 8 shows the comparative cost analysis of the system components in terms of the NPC and COE for the three HESs. The NPC and COE are calculated for the entire system at the expected life of 25 years. The NPC and COE are considered to be an appropriate indicator to appraise the optimal plan for the RE system as compared to the non-RE system (Acar and Dincer 2014). The COE is an essential economic criterion for optimizing hybridized electrical systems, which is the mean cost per kilowatt hour (\$/KWh) of useful electric energy.

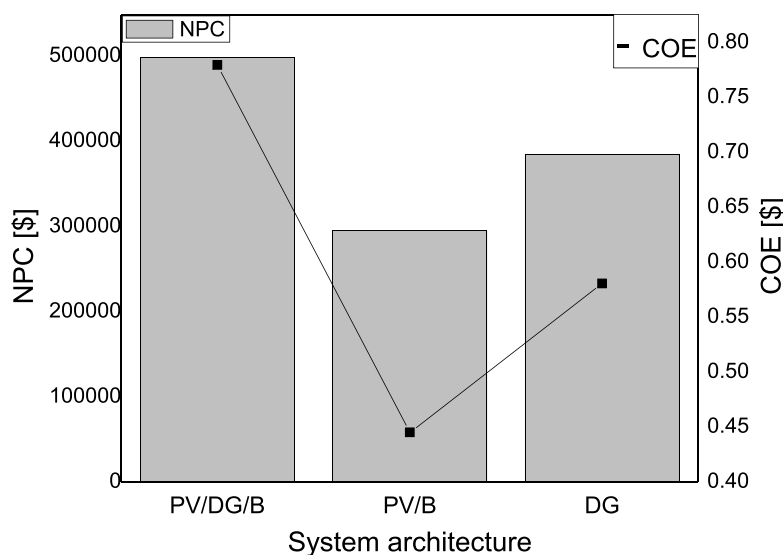
Figure 9, the pure RE-based system has the lowest NPC and COE, followed by the DG and the RE-based systems. Although the DG system has a lower capital cost compared to the PV/DG/B, the amount of carbon emission from the DG-system is considerably high, compared to the other HESs. The lowest NPC and COE values for the PV system can be explained by the low cost of PV price and can be ascribed to the direct relationship between solar radiations and the NPC and COE. The higher the solar radiation, the lower the NPC and COE values. The PV/B system has NPC and COE values of

\$294,777.10 and \$0.4450/KWh, respectively, compared to the PV/DG/B system that has NPC and COE values of \$498,386.2 and \$0.7794/KWh, respectively, making the PV/B HES the preferred choice to meet the Umuejim Amorka community's electricity demand.

4.3. System emission and diesel consumption

The total evaluation of the pollutant gas produced by various HESs is shown in Table 5. As expected, the annual rate of carbon emission and other related particles experienced in the hybrid energy system is associated with the number of liters of diesel consumed by the generator each year (Szabo et al. 2011). Table 5 shows an assessment of the emission of the three HESs considered in this study.

While the DG-based HESs display different levels of emission particles, the RE systems show no emission (Table 5). The PV/DG/B and PV/B HES configurations provide greater savings in terms of CO₂, CO, and SO₂ emission, compared to the DG system. The utilization of PV and battery banks in the PV/DG/B and PV/B HESs enhanced the reduction of carbon emission by 2.4% and 100%, respectively. Although the RE-based HES (PV/B) considered in this study displayed evidence of no emission, the lowest NPC and COE values observed in the PV/B HES compared with PV/DG/B and DG HESs may provide a better choice for application of the PV/B system to meet the electrical load demand in the Umuejim Amorka community.

**Figure 8.**

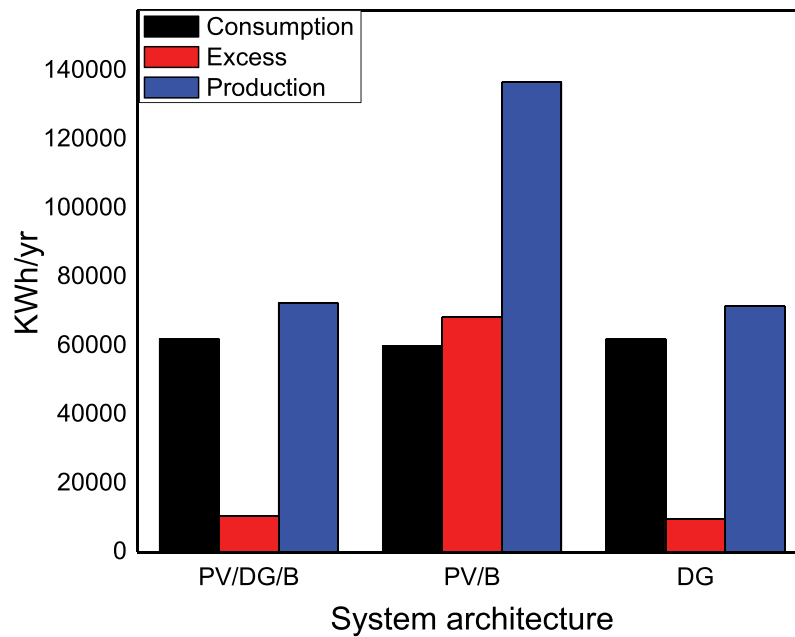


Figure 9. System cost comparison in terms of the NPC and COE.

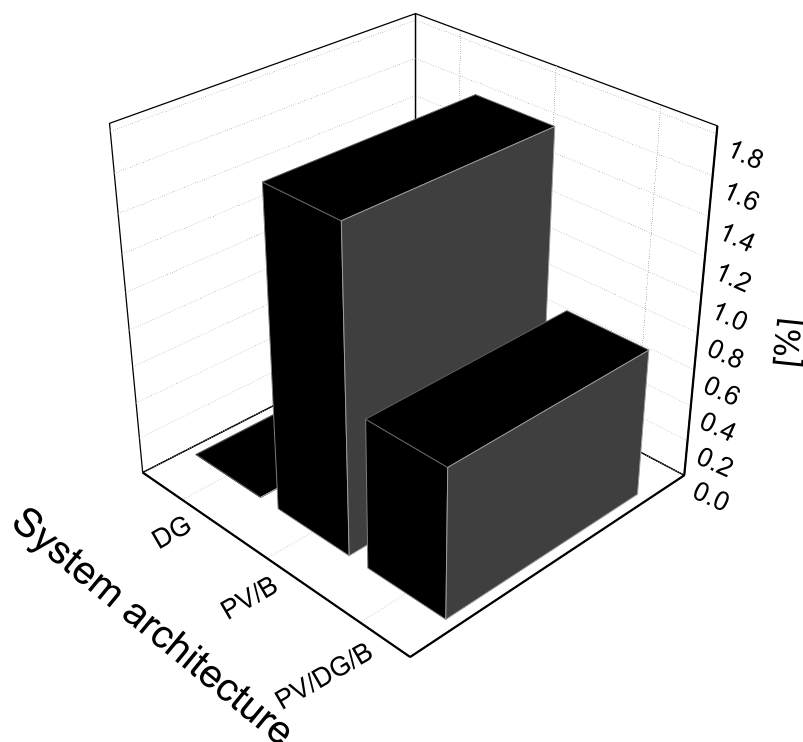


Figure 10. Comparison of energy production, consumption, and excess by the three HESs.

4.4. Energy parametric study comparison

The comparison of electric generation, consumption, and excess of the three HESs is presented in Figure 10. The statistics show that the PV/DG/B, PV/B and DG HESs generate total electricity of 72,482 KWh/year, 136,673 KWh/year, and 71,675 KWh/year, respectively, which are used to charge the battery and supply the electrical load.

Figure 10 shows that while the two diesel-based HESs (PV/DG/B, DG) have an equal electric energy consumption of 62,050 KWh/year, the RE-based HES (PV/B) has an electricity consumption of 59,903 KWh/year. The similarity in the energy consumption can be attributed to the average annual load demand in the community systems. The RE-based HES produced the highest excess electric energy of 68,384 KWh/year,

Table 5. The output from the economic and environmental parameters for the three HESs.

Emissions (Kg yr ⁻¹)						
System configuration	CO ₂ (Kg/yr)	CO (Kg/yr)	Unburned hydrocarbon (Kg/yr)	Particulate matter (Kg/yr)	SO ₂ (Kg/yr)	NO ₂ (Kg/yr)
PV/DG/B	69062	435	19.0	2.64	169	409
PV/B	0.0	0.0	0.0	0.0	0.0	0.0
DG	70829	446	19.5	2.71	173	419

and the diesel-based HESs (PV/DG/B, DG) displayed excess electricity of 10,432 KWh/year and 9,625 KWh/year respectively. When the electric power produced by PV modules exceeds the load demand, the excess electric power is stored in the batteries for use during low supply from the other electric sources.

4.5. Return on Investment (ROI) analysis

The economic viability of the investment by the Umuejim Amorka community in the electrification project requires further analysis. Economic parameters such as ROI are often performed to understand the feasibility and sustainability of the project. Return on Investment analysis is used to process the gain or loss produced on investment compared to the amount of money financed. Figure 10 shows the ROI results of the three HESs under consideration.

As earlier pointed out, the selection of the most efficient and economical HES is necessary for the community to decide on the HES to employ in the community rural electrification. As observed in Figure 10, while the other two HESs have different ROI values, the PV/B system has the highest ROI values of 1%, making the system the best option for the Umuejim Amorka community electrification. The highest ROI for the PV/B system can be ascribed to the efficiency of the system and the potential of providing profit throughout the project life cycle.

5. Conclusion

A cost-effective modeling technology was used to study the feasibility of three different HESs to meet the high electric energy demand of an off-grid community. Through assessment of the available resources prevailing in the community, environmental and economic impacts of the resources were simulated and analyzed using HOMER software. Measured household load and system components' cost were used in the simulation. Our findings suggest that while pure RE-based HES (PV/B) has the lowest NPC and COE compared to the diesel-based HESs (PV/DG/B, DG), the RE-based systems were found to fulfill the electric demand with no emission compared to the emission diesel-based system (DG) that is currently used in rural communities. Further analysis and comparison of the three HESs in terms of capital and total cost revealed that the PV/B HES has the lowest capital and total costs compared to the other HESs. The results also

showed that apart from the DG-based high emission system, the RE-based system has the highest ROI value, making the PV/B system the preferred choice to fulfill the community electricity demand at a lower cost. Furthermore, this study has highlighted the importance of RE-based projects in rural communities through the improvement of socio-economic developments and job provision for residents.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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